

Predicting Cross-Sectional Relative Returns in the S&P 500: A Feature-Engineering and Walk-Forward Modeling Case Study

Quantitative Research Case Study

1 Problem and Approach

The task is to predict *relative* future stock returns inside the S&P 500: not whether the market goes up, but which stocks will out- or under-perform their peers. This framing is deliberate. Index-level direction is dominated by macro factors that daily OHLCV data cannot forecast, whereas the *cross-section* of returns carries persistent, tradable structure (momentum, reversal, liquidity, volatility, and sector-rotation effects) that is learnable from price and volume alone.

We therefore predict each stock’s forward return *in excess of its market and sector*, rank stocks by the model score on each day, and form a dollar-neutral long–short portfolio (long the top decile, short the bottom decile). The entire pipeline is built to be honest about look-ahead leakage, overlapping-window inflation, and survivorship bias, because in this problem the easiest way to produce a spectacular-looking result is to make a methodological mistake.

Headline design. Curated technical/liquidity/cross-sectional features → sector- and market-relative forward-return targets → a single-factor momentum baseline plus Ridge, LightGBM, XGBoost, and a small MLP → expanding yearly walk-forward with a final untouched hold-out → rank IC / ICIR and a transaction-cost-aware long–short backtest. The headline horizon is the 5-trading-day relative return.

2 Data

We use only two files: `sp500_stocks.csv` (daily OHLCV per ticker) and `sp500_companies.csv` (sector, sub-industry, headquarters, founding year, and index `date_added`). No fundamentals, analyst, macro, or external listing data are used. The panel covers ≈ 503 symbols from 2000-01 to 2026-06 ($\approx 2.9\text{M}$ stock-day rows).

Price adjustment (data quality). The raw file exposes only a `close` column. We verified it is already *split-adjusted*: around NVDA’s 2024-06-10 10:1 split ($120.68 \rightarrow 121.58$) and AAPL’s 2020-08-31 4:1 split ($121.06 \rightarrow 125.17$) there is no split-day gap, consistent with a `yfinance auto_adjust=True` export. All returns, momentum, volatility, and targets are thus computed on an adjusted price, so splits do not inject artificial jumps. Residual uncertainty about dividend adjustment is treated as a minor data-quality caveat.

Survivorship bias. The constituent list is effectively the *current* membership applied to history, so firms that went bankrupt, were delisted, or were acquired (Enron, Lehman, Bear Stearns, WaMu, Sears, Kodak, Yahoo, Celgene, Xilinx, ...) are absent. This biases backtests optimistically and is discussed quantitatively in §6. We do not claim to correct it; we bound its impact and avoid designs that amplify it.

3 Feature Engineering

Each row is one stock-day observation, and every feature uses only information available at or before that date (all rolling windows are backward-looking with an explicit `shift`). We build ≈ 500 features in economically motivated groups:

- **Price / momentum / reversal:** multi-horizon returns (1–252d), log returns, skip-window momentum, momentum acceleration, short-term reversal, intraday/overnight decomposition, close-in-bar location. *Captures short-term reversal and medium-term momentum.*
- **Volatility / tail / drawdown:** rolling, upside/downside, Parkinson and Garman–Klass volatility, ATR, skew/kurtosis, drawdown from recent highs. *Controls for risk state and crash sensitivity.*
- **Trend / technical:** SMA/EMA gaps and slopes, MACD, RSI, Bollinger z-score/bandwidth/%b, channel position, breakout distance, Williams %R, CCI. *Encodes overbought/oversold and trend regimes.*
- **Liquidity / volume:** log dollar volume, volume z-scores and momentum, Amihud illiquidity, return-per-dollar-volume. *Captures attention, liquidity shocks, and transaction-cost pressure.*
- **Market regime:** equal-weight market return, breadth, cross-sectional dispersion, market volatility. *Relative-return predictability is regime-dependent.*
- **Sector / sub-industry and peer / style / geography:** group returns, breadth, dispersion, leave-one-out peer means, headquarters-region peers, and style buckets. *Separates stock-specific alpha from group movement.*
- **Cross-sectional transforms:** date-wise z-scores/ranks plus sector- and sub-industry-neutral versions of the key signals. *Because the target is relative, cross-sectional normalization is usually more informative than raw levels.*

Prediction target. For horizon h we define the forward total return $r_{i,t}^{(h)}$ and subtract a contemporaneous group mean to obtain sector/market-relative targets, e.g. `target_excess_sector_fwd_hdi,t` = $r_{i,t}^{(h)} - \bar{r}_{\text{sector}(i),t}^{(h)}$. Targets are forward-looking and are strictly excluded from the feature matrix. We evaluate horizons $h \in \{1, 5, 20, 30, 90\}$ with $h=5$ as the headline.

4 Methodology

4.1 Models

We compare, from simple to flexible: (i) a **single-factor momentum** baseline (rank by 12-minus-1-month momentum); (ii) **Ridge** regression (linear, strong regularization); (iii) **LightGBM** and (iv) **XGBoost** gradient-boosted trees (non-linear, interaction-aware); and (v) a small **PyTorch MLP** (two hidden layers, dropout, weight decay, early stopping on validation). The MLP is kept deliberately small and is *not* tuned until it wins; §5 reports its honest standing — competitive on in-sample validation IC but, like every trained model here, unable to beat a single momentum factor out of sample.

4.2 Leakage controls

Every result below is produced under the same defenses:

- **One-day execution lag:** scores computed on day t are traded at $t+1$, so no same-bar information

is used.

- **Label purge:** training rows whose forward label window crosses a split boundary are dropped, on both the training and the evaluation side.
- **Horizon embargo:** the validation/test window starts an h -day embargo after training ends, removing label overlap at the seam.
- **Train-only fitting:** all preprocessing statistics (imputation, scaling, winsorization reference) and the model are fit on training rows only.
- **Universe & winsorization:** each day we drop the bottom 10% by dollar volume and winsorize features at the 1%/99% level, so illiquid names and outliers do not drive the fit or an untradeable backtest.

4.3 Avoiding overlapping-window inflation

A subtle but decisive issue: if an h -day forward return is booked every day, consecutive observations overlap by $h-1$ days, their IC series is strongly autocorrelated, and a naive $ICIR = \sqrt{252} \cdot \overline{IC} / \sigma(IC)$ together with an overlap-booked Sharpe explode with h — a pure artifact, not alpha. We fix this in two ways: (1) the portfolio rebalances every h days so booked returns are *non-overlapping*; and (2) ICIR is annualized from a non-overlapping IC subsample (one observation per h days, scaled by $\sqrt{252/h}$). We retain the naive value as `rank_ic_ir_raw` only to document the inflation (Fig. 3). A sanity gate flags any run with Sharpe > 3 or ICIR > 5 as suspected leakage/overlap.

4.4 Validation: expanding walk-forward + hold-out

A single static split can land in a lucky regime. We instead use **expanding yearly walk-forward**: train on all data up to year Y , validate on year $Y+1$ (with purge + embargo), then roll forward one year and expand the training window. Model selection uses the *across-fold mean validation ICIR* only. A final post-2022 **hold-out** is evaluated exactly once, after selection, and never used for tuning.

4.5 Portfolio and metrics

Daily scores become a top-decile long / bottom-decile short, dollar-neutral portfolio rebalanced every h days, with a 5 bps per-side transaction cost applied to turnover. We report rank IC, overlap-adjusted ICIR, decile spread, annualized net return, volatility, net Sharpe, max drawdown, and turnover.

5 Results

Validation selection. Selecting by across-fold mean validation ICIR, the small MLP is the strongest model at the 1-, 5-, and 30-day horizons (e.g. sector-relative 5d ICIR 1.15, market-relative 0.99) and LightGBM at 20 days (market-relative 0.99); Ridge and XGBoost trail (Fig. 2). No single model dominates every horizon and the boosted trees do *not* uniformly win — the MLP is genuinely competitive on the in-sample IC metric, so we do not pretend trees are categorically better.

The decisive test: the untouched hold-out. The validation ranking does not survive out of sample. On the 2022–2026 hold-out the *single-factor momentum baseline* (rank by 12-minus-1-month momentum, no fitting) beats every trained multivariate model at the headline 5-day horizon —

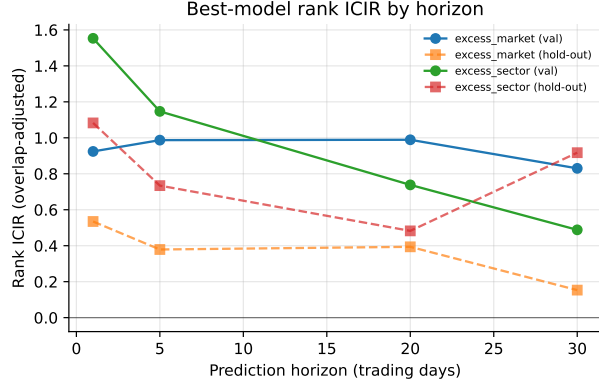


Figure 1: Best-model rank ICIR by horizon (validation vs hold-out), per target family.

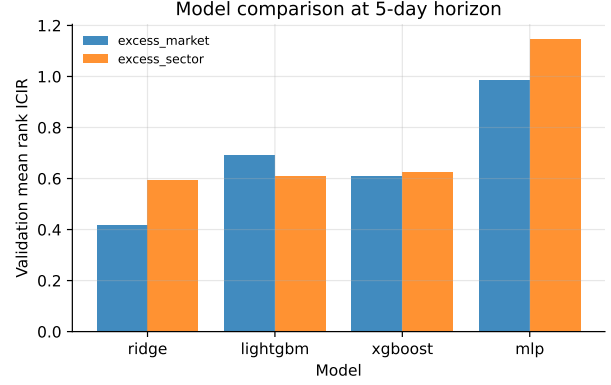


Figure 2: Validation mean ICIR by model at the 5-day horizon.

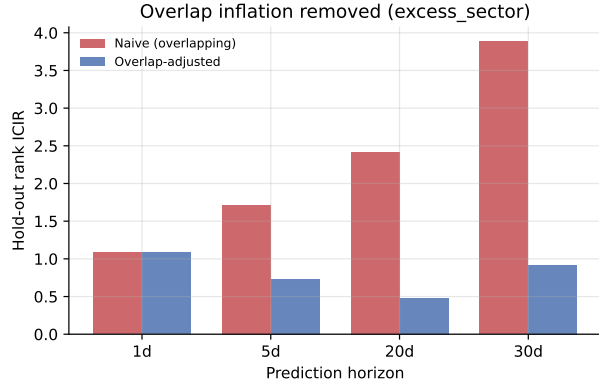


Figure 3: Overlap-adjusted vs naive (overlapping) hold-out ICIR by horizon.

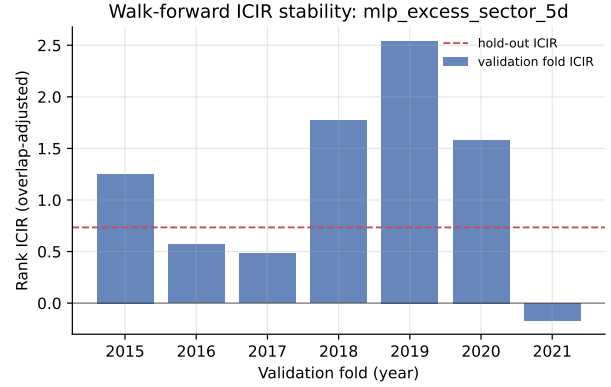


Figure 4: Walk-forward per-fold ICIR for the headline run.

higher rank IC (0.024 vs ≤ 0.013), higher ICIR (1.06 vs ≤ 0.73), higher net Sharpe (1.05 vs ≤ 0.53), and roughly one-quarter the turnover (0.16 vs ~ 0.59); the same ordering holds at 20 days (Table 1). The models’ validation-IC edge is therefore largely overfit: a robust, low-turnover factor is hard to beat after costs. This is the central, deliberately humbling finding of the study.

Overlap correction matters. Fig. 3 contrasts the overlap-adjusted hold-out ICIR with the naive overlapping value. The naive metric inflates sharply with horizon (to ≈ 3 –10 at 30–90 days) while the adjusted metric stays modest (≤ 1.3); reporting the naive number would manufacture a long-horizon “signal” that is pure window overlap. The Sharpe $> 3/\text{ICIR} > 5$ sanity gate flags none of the adjusted runs.

Stability across regimes. Fig. 4 shows per-fold ICIR across the expanding walk-forward. The signal is positive but *variable* across folds, and the hold-out value (dashed) sits below the validation mean — the expected out-of-sample decay. The 90-day horizon is excluded from selection (too few non-overlapping validation periods for a stable ICIR), and the longer-horizon hold-out Sharpe values are estimated from few non-overlapping periods and should be read as noisy.

Hold-out performance. Table 1 reports the untouched post-2022 hold-out, net of 5 bps/side cost. The numbers are deliberately modest and the simple baseline leads: realistic cross-sectional equity “alpha” from price/volume features alone is small, and the trained models’ apparent in-sample edge

Table 1: Untouched hold-out (2022–2026), net of 5 bps/side cost. The single-factor momentum baseline beats the validation-selected models on IC, ICIR, Sharpe, return, *and* turnover. Parenthetical ICIR is the naive overlapping value, shown only to document inflation. Best per block in **bold**.

Horizon	Config	Rank IC	ICIR (raw)	Net Sharpe	Ann. Ret	Turnover
5d	Momentum baseline (sector)	0.024	1.06 (2.10)	1.05	13.5%	0.16
5d	MLP (<i>val-selected, sector</i>)	0.013	0.73 (1.72)	0.17	1.2%	0.59
5d	Ridge (sector)	0.010	0.40 (0.99)	0.53	5.1%	0.65
5d	LightGBM (sector)	0.009	0.20 (1.17)	0.24	1.7%	0.53
20d	Momentum baseline (market)	0.043	0.71 (3.62)	1.12	13.1%	0.29
20d	LightGBM (<i>val-selected, market</i>)	0.028	0.39 (2.47)	0.49	5.5%	0.64
20d	Ridge (market)	0.029	0.54 (2.27)	0.76	9.8%	0.61

does not translate into out-of-sample value here.

6 Discussion

Overfitting and the validation–hold-out gap. The clearest evidence of overfitting is in the results themselves: models selected on validation ICIR fail to beat a single momentum factor out of sample, and the model with the best validation ICIR (the MLP) is not the best on the hold-out (Ridge and the baseline trade better). We use heavy regularization (Ridge penalty, tree depth/leaf/subsample limits, MLP dropout + weight decay + early stopping), select hyperparameters on validation folds only, and report a single untouched hold-out precisely so this gap is *visible* rather than hidden, with the Sharpe > 3/ICIR > 5 gate catching anything “too good”.

Look-ahead leakage. Addressed structurally: one-day execution lag, label purge on both sides, horizon embargo, and train-only preprocessing. We also verified that a no-embargo diagnostic inflates metrics, confirming the controls bind.

Survivorship bias. The dataset is current-membership-on-history, so results are optimistically biased. The long–short construction partially offsets this (the missing names would populate *both* legs), but the short book of historically-distressed firms is under-represented, so realized short alpha would likely be larger and the long book’s survivorship premium smaller. We treat the reported hold-out as an upper bound and avoid long-horizon buy-and-hold of survivors, which would amplify the bias most.

Data quality. Prices are split-adjusted; dividend adjustment is uncertain and treated as a minor caveat. We winsorize features and filter the illiquid decile so bad ticks and microcaps do not dominate.

Turnover and market impact. We charge 5 bps per side on realized turnover and rebalance only every h days. Turnover is itself decisive here: the trained models churn $\sim 60\%$ per rebalance versus $\sim 16\%$ for the slow momentum factor (Table 1), so even where a model’s gross IC is comparable its net value is eroded. Market impact beyond the linear cost is not modeled, and capacity is limited by the bottom-decile short book — the binding real-world constraint.

Regime dependence. Walk-forward folds span 2008, 2020, and 2022 stress periods; Fig. 4 shows the signal is not uniform across regimes, motivating the market/breadth/dispersion regime features rather than a single static model.

Economic intuition. The one signal that clearly survives out of sample is cross-sectional momentum

— a well-documented behavioural underreaction effect — which is precisely why the single-factor baseline is so hard to beat. The multivariate models mostly add estimation variance, not orthogonal signal, on top of it.

7 Limitations and Conclusion

This study delivers an honest, reproducible cross-sectional relative-return pipeline for the S&P 500 using only price/volume and light metadata. The key discipline is methodological: non-overlapping evaluation, walk-forward selection, explicit leakage controls, and a single-factor baseline that the models must – but fail to – beat. The realistic, useful conclusion is sobering: with these features a carefully regularized multivariate model does *not* reliably beat a single robust momentum factor on an untouched hold-out, despite a higher in-sample IC, while carrying far higher turnover. The baseline’s edge is also partly a 2022–2026 momentum-regime effect, which is itself the argument for the regime features and for never over-trusting one hold-out. The highest-value next steps are point-in-time constituents to remove survivorship bias, fundamental and alternative data to add genuinely orthogonal signal, and explicit capacity/impact modeling. (Graph-embedding and foundation-model experiments exist in the repository but are out of scope for this core study.)